

DAX CALCULATIONS

Practical No.:	PL-300 Practical 4
Practical Title:	Create DAX Calculations in Power BI Desktop
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Project Summary

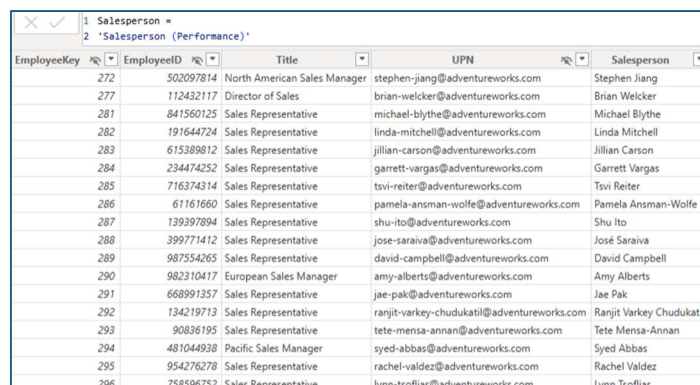
Created calculated tables, calculated columns, and simple measures using Data Analysis Expressions (DAX). Met the objective to create calculated tables, create calculated columns, create measures. A Matrix visual was configured to display these date-level summaries, and target metrics were enhanced with Variance and Variance Margin measures.

Objectives

Calculated tables and columns	Measures
Create the Salesperson calculated table Create the Date table and added calculated columns while making sure the Date table is complete	Create simple measures Create additional measures

Overview

1. Created a Salesperson table
 - a. Salesperson = 'Salesperson (Performance)'
 - b. Created a relationship between Salesperson and Sales using EmployeeKey
 - c. Hide columns: EmployeeID, EmployeeKey, UPN



1 Salesperson =				
2 'Salesperson (Performance)'				
EmployeeKey	EmployeeID	Title	UPN	Salesperson
272	502097814	North American Sales Manager	stephen-jiang@adventureworks.com	Stephen Jiang
277	112432117	Director of Sales	brian-welcker@adventureworks.com	Brian Welcker
281	841560125	Sales Representative	michael-blythe@adventureworks.com	Michael Blythe
282	191644724	Sales Representative	linda-mitchell@adventureworks.com	Linda Mitchell
283	615389812	Sales Representative	jillian-carson@adventureworks.com	Jillian Carson
284	234474252	Sales Representative	garrett-vargas@adventureworks.com	Garrett Vargas
285	716374314	Sales Representative	tsvi-reiter@adventureworks.com	Tsvi Reiter
286	61161660	Sales Representative	pamela-ansman-wolfe@adventureworks.com	Pamela Ansman-Wolfe
287	139397894	Sales Representative	shu-ito@adventureworks.com	Shu Ito
288	399771412	Sales Representative	jose-saraiva@adventureworks.com	José Saraiva
289	987554265	Sales Representative	david-campbell@adventureworks.com	David Campbell
290	982310417	European Sales Manager	amy-alberts@adventureworks.com	Amy Alberts
291	668991357	Sales Representative	jae-pak@adventureworks.com	Jae Pak
292	134219713	Sales Representative	ranjit-varkey-chudukatil@adventureworks.com	Ranjit Varkey Chudukatil
293	90836195	Sales Representative	tete-mensa-annan@adventureworks.com	Tete Mensa-Annan
294	481044938	Pacific Sales Manager	syed-abbas@adventureworks.com	Syed Abbas
295	954276278	Sales Representative	rachel-valdez@adventureworks.com	Rachel Valdez
296	758596752	Sales Representative	lynn-tsolfias@adventureworks.com	Lynn Tsolfias

Figure 1: Final Salesperson table

2. Created a Date table
 - a. Date = CALENDARAUTO(6)
 - b. Formatting used was (yyyy/mm/dd)
 - c. Added columns: Year, Quarter and month

- d. Grouping was done for
 - i. Fiscal (year, month and quarter),
 - ii. Pricing (Avg, Max, Min, Median Price) and,
 - iii. Counts (Orders, Order Lines)

1 Date = CALENDARAUTO(6)

Date	Year	Quarter	Month	MonthKey
2017/07/01 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/02 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/03 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/04 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/05 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/06 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/07 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/08 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/09 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/10 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/11 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707
2017/07/12 00:00:00	FY2018	FY2018 Q1	2017 Jul	201707

Figure 2: Final Date table

3. Added Martix visual type that included:
 - a. Year, Avg, Median, Min, Max, Orders, order lines

Year	Avg Price	Median Price	Min Price	Max Price	Orders	Order Lines
FY2018	R748,68	R419,46	R4,75	R2 146,96	739	8 459
2017 Jul	R655,59	R419,46	R5,19	R2 146,96	38	352
2017 Aug	R758,93	R419,46	R4,75	R2 146,96	75	785
2017 Sep	R741,85	R419,46	R5,19	R2 146,96	60	593
2017 Oct	R677,45	R419,46	R5,19	R2 146,96	40	499
2017 Nov	R752,31	R419,46	R5,01	R2 146,96	90	1 106
2017 Dec	R734,58	R419,46	R5,01	R2 146,96	63	803
2018 Jan	R808,94	R419,46	R5,19	R2 146,96	40	377
2018 Feb	R896,80	R419,46	R5,01	R2 146,96	79	866
2018 Mar	R863,54	R419,46	R5,19	R2 146,96	64	653
2018 Apr	R732,25	R419,46	R5,19	R2 146,96	37	494
2018 May	R761,30	R419,46	R4,75	R2 146,96	85	1 112
2018 Jun	R552,95	R419,46	R4,75	R2 146,96	68	819
FY2019	R397,81	R202,33	R4,32	R1 466,01	1 255	21 670
2018 Jul	R350,74	R196,33	R4,75	R1 466,01	72	1 723
2018 Aug	R372,75	R202,33	R4,75	R1 466,01	139	2 964
2018 Sep	R365,30	R196,33	R4,32	R1 466,01	111	2 185
2018 Oct	R382,00	R202,33	R5,19	R1 466,01	73	1 406
2018 Nov	R412,48	R209,26	R5,01	R1 466,01	133	2 345
2018 Dec	R407,08	R202,33	R4,75	R1 466,01	114	1 732
2019 Jan	R428,87	R324,45	R5,19	R1 466,01	65	983
2019 Feb	R488,75	R469,79	R5,19	R1 466,01	132	1 664
2019 Mar	R450,03	R324,45	R5,01	R1 466,01	106	1 215
2019 Apr	R369,78	R198,04	R5,01	R1 466,01	74	1 449
2019 May	R400,51	R202,33	R4,75	R1 466,01	134	2 306
Total	R446,39	R214,24	R1,33	R2 146,96	3 616	57 851

Figure 2: Visual Matrix of the Sales table

4. Editing for the Target table was done by
 - a. Adding Variance and Variance Margin measures
 - b. Editing the table on report view

Salesperson	Sum of Sales	Target	Variance	Variance Margin
Amy Alberts	\$10 288 626	R19 450 000	-R9 161 374	-47,10%
Brian Welcker	\$77 548 570	R221 700 000	-R144 151 430	-65,02%
David Campbell	\$12 004 822	R19 625 000	-R7 620 178	-38,83%
Garrett Vargas	\$13 875 633	R23 675 000	-R9 799 367	-41,39%
Jae Pak	\$8 410 883	R13 575 000	-R5 164 117	-38,04%
Jillian Carson	\$7 633 387	R13 675 000	-R6 041 613	-44,18%
José Saraiva	\$13 875 633	R18 875 000	-R4 999 367	-26,49%
Linda Mitchell	\$25 634 503	R40 850 000	-R15 215 497	-37,25%
Lynn Tsofilas	\$1 391 025	R3 210 000	-R1 818 975	-56,67%
Michael Blythe	\$21 987 348	R31 150 000	-R9 162 652	-29,41%
Pamela Ansman-Wolfe	\$30 005 939	R53 850 000	-R23 844 061	-44,28%
Rachel Valdez	\$1 877 743	R4 125 000	-R2 247 257	-54,48%
Ranjit Varkey Chudukatil	\$4 527 840	R9 050 000	-R4 522 160	-49,97%
Shu Ito	\$18 001 116	R59 850 000	-R41 848 884	-69,92%
Stephen Jiang	\$65 868 919	R110 150 000	-R44 281 081	-40,20%
Syed Abbas	\$1 391 025	R3 050 000	-R1 658 975	-54,39%
Tete Mensa-Annan	\$12 004 822	R17 100 000	-R5 095 178	-29,80%
Tsvi Reiter	\$7 638 607	R13 250 000	-R5 611 393	-42,35%
Total	\$77 548 570			

Figure 4: Visuals for the Target table

Images

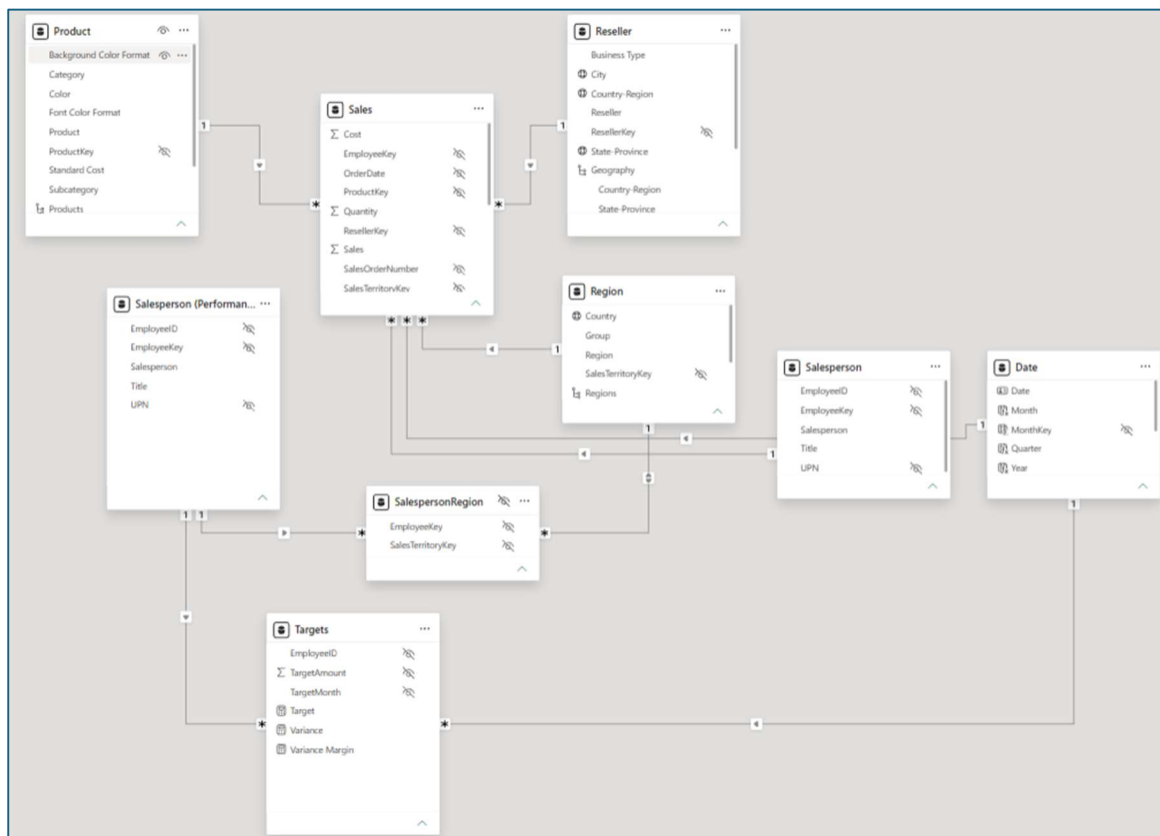


Figure 5: Final Model View

Notes

The target column looks different from the one displayed in the example due to the formatting of the decimals

Salesperson	Sales	Target
Amy Alberts	\$10,288,626	\$19,450,000
Brian Welcker	\$77,548,570	\$221,700,000
David Campbell	\$12,004,822	\$19,625,000
Garrett Vargas	\$13,875,633	\$23,675,000
Jae Pak	\$8,410,883	\$13,575,000
Jillian Carson	\$7,633,387	\$13,675,000
José Saraiva	\$13,875,633	\$18,875,000
Linda Mitchell	\$25,634,503	\$40,850,000
Lynn Tsoflias	\$1,391,025	\$3,210,000
Michael Blythe	\$21,987,348	\$31,150,000
Pamela Ansman-Wolfe	\$30,005,939	\$53,850,000
Total	\$77,548,570	

Figure 6a: Target View from example

Salesperson	Sum of Sales	Target
Amy Alberts	\$10 288 626	R19 450 000
Brian Welcker	\$77 548 570	R221 700 000
David Campbell	\$12 004 822	R19 625 000
Garrett Vargas	\$13 875 633	R23 675 000
Jae Pak	\$8 410 883	R13 575 000
Jillian Carson	\$7 633 387	R13 675 000
José Saraiva	\$13 875 633	R18 875 000
Linda Mitchell	\$25 634 503	R40 850 000
Lynn Tsoflias	\$1 391 025	R3 210 000
Michael Blythe	\$21 987 348	R31 150 000
Pamela Ansman-Wolfe	\$30 005 939	R53 850 000
Total	\$77 548 570	

Figure 7a: My target view

Formulas

- Salesperson ='Salesperson (Performance)'
- Date =CALENDARAUTO(6)
- Year ="FY" & YEAR('Date'[Date]) + IF(MONTH('Date'[Date]) > 6, 1)
- Month =FORMAT('Date'[Date], "yyyy MMM")
- MonthKey =(YEAR('Date'[Date]) * 100) + MONTH('Date'[Date])
- Avg Price =AVERAGE(Sales[Unit Price])
- Median Price =MEDIAN(Sales[Unit Price])
- Min Price =MIN(Sales[Unit Price])
- Max Price =MAX(Sales[Unit Price])
- Orders =DISTINCTCOUNT(Sales[SalesOrderNumber])
- Order Lines =COUNTROWS(Sales)
- Variance Margin =DIVIDE([Variance], [Target])

More formulas on the assets folder